

Distributional Divergence.

objective : Given a pair of distribution functions defined on the same sample space, define / quantify "distance" b/w them.

Given an event $A \in \mathcal{F}$,

define $I(A) \triangleq -\log P(A)$

suppose x is a DRV, then the "surprisal" associated with it is quantified as follows:

$$\mathbb{E}_{P_x} [-\log P(x_i)]$$

$$\mathbb{E}_{P_x} f(x) = \sum P_x \cdot f(x)$$

Entropy $\hookrightarrow$ $H(P_x) = \sum_i -P(x_i) \log P(x_i)$: avg information in P_x

suppose P_x & Q_x are two distributions on the same sample space.

Define

$$\mathbb{E}_{P_x} -\log Q(x) = H(P, Q)$$

↳ cross-entropy.

Consider

$$-H(P) + H(P, Q)$$

$$+ \sum P(x) \log P(x) - \sum P_x \log Q(x)$$

$$D_{KL}(P \parallel Q) = \sum P(x) \log \frac{P(x)}{Q(x)}$$

KL-divergence.

$$\text{To show: } D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$$

$$D_{KL} \geq 0$$

$$D_{KL} = 0 \quad \text{iff} \quad P = Q$$

$$D_{KL}(P_x \parallel Q_x) = \int_x P_x(x) \log \frac{P_x(x)}{Q_x(x)} dx.$$

Suppose we have N samples drawn iid

$$D = \{v_1, v_2, v_3, \dots, v_N\} \sim \text{iid } p_v$$

$v_i \in \mathbb{R}^d$

Goal : Estimate p_v , given D .

Assume $p_\theta(v)$ as the model density

compute $D_{KL}(p_v \parallel p_\theta) = \int p_v(v) \log \frac{p_v(v)}{p_\theta(v)} dv$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} D_{KL}(p_v \parallel p_\theta)$$

$$= \underset{\theta}{\operatorname{argmin}} \left[\int p_v(v) \log p_v(v) dv - \int p_v(v) \log p_\theta(v) dv \right]$$

Independent of θ .

$$= \underset{\theta}{\operatorname{argmin}} - \int p_v(v) \log p_\theta(v) dv$$

$$= \underset{\theta}{\operatorname{argmin}} - \mathbb{E}_{p_v} \log p_\theta(v)$$

from the Law of Large numbers,

$$\mathbb{E}_{p_v} \log p_\theta(v) \approx \frac{1}{N} \sum_{i=1}^N \log p_\theta(v_i)$$

where $v_i \sim \text{iid } p_v$.

$$\frac{1}{N} \sum_{i=1}^N \log p_\theta(v_i) \longrightarrow \mathbb{E}_{p_v} \log p_\theta(v) \text{ as } N \rightarrow \infty.$$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left[-\frac{1}{N} \sum_{i=1}^N \log p_\theta(v_i) \right]$$

$v_i \sim \text{iid } p_v$.

$\equiv$ minimal DKL estimator.

$$= \underset{\theta}{\operatorname{argmax}} \left[\frac{1}{N} \sum_{i=1}^N \log p_\theta(v_i) \right]$$

$p_\theta(v_i) \equiv$ likelihood of v_i under p_θ .

$\equiv$ maximum likelihood estimator.

Joint likelihood of the data

$$l_{\theta}(D) = \prod_{i=1}^N p_{\theta}(v_i) \quad \text{"." iid}$$

$$\hat{l}_{\theta}(D) = \sum_{i=1}^N \log p_{\theta}(v_i)$$

$$\theta^* = \operatorname{argmax}_{\theta} \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(v_i).$$

How to solve the ML Estimation problem in practice?

a) How to choose p_{θ} ?

b) How to solve the optimization problem?

a) p_{θ} is chosen using a canonical function approximator.

Eg: Assume $p_{\theta}(\cdot) \sim N(\cdot; h_{\theta}(v), I)$

$h_{\theta}(v)$: hypothesis function / Model.

b) optimization is mostly numerical.

Consider the problem of supervised regression.

$$x \in \mathbb{R}^d, \quad y \in \mathbb{R}$$

$$\text{Data } D = \left\{ (x_i, y_i) \right\}_{i=1}^N \sim \text{iid } p_{xy}$$

Estimate $p_{y|x}$

$$p_{\theta}(y|x) = N(y; h_{\theta}(x), I) : \underline{\text{Model}}$$

$$h_{\theta}(x) : x \rightarrow \mathbb{R}$$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \frac{-1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | x_i)$$

$$\propto \underset{\theta}{\operatorname{argmin}} - \sum_{i=1}^N \log \exp \left[(y_i - h_{\theta}(x_i))^2 \right]$$

$$\theta^* \propto \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^N [y_i - h_{\theta}(x_i)]^2$$